Design of Shift System

Design a digital system with three 4-bit registers, A, B, and C, to perform the following operations:

1. Transfer two binary numbers to A and B when a start signal is enabled.
2. If $A < B$, shift left the contents of A and transfer the result to register C.
3. If $A > B$, shift right the contents of B and transfer the result to register C.
4. If $A = B$, transfer the number to register C unchanged.

PIN ASSIGNMENT:

(Intel Quartus Prime)

	tatu	From	To	Assignment Name	Value	Enabled	Entity	Comment	Tag
501	?		HSMC...T_N2	Location	PIN_V24	Yes			
502	?		HSMC...T_N1	Location	PIN_G24	Yes			
503	✓		in clk	Location	PIN_Y2	Yes			
504	✓		in start	Location	PIN_M21	Yes			
505	✓		in reset	Location	PIN_M23	Yes			
506	✓		out C[3]	Location	PIN_H16	Yes			
507	✓		out C[2]	Location	PIN_J15	Yes			
508	✓		out C[1]	Location	PIN_G17	Yes			
509	✓		out C[0]	Location	PIN_J17	Yes			
510	✓		out B[3]	Location	PIN_H19	Yes			
511	✓		out B[2]	Location	PIN_J19	Yes			
512	✓		out B[1]	Location	PIN_E18	Yes			
513	✓		out B[0]	Location	PIN_F18	Yes			
514	✓		out A[3]	Location	PIN_F21	Yes			
515	✓		out A[2]	Location	PIN_E19	Yes			
516	✓		out A[1]	Location	PIN_F19	Yes			
517	✓		out A[0]	Location	PIN_G19	Yes			
518	✓		in B_in[3]	Location	PIN_AB26	Yes			
519	✓		in B_in[2]	Location	PIN_AD26	Yes			
520	✓		in B_in[1]	Location	PIN_AC26	Yes			
521	✓		in B_in[0]	Location	PIN_AB27	Yes			
522	✓		in A_in[3]	Location	PIN_AD27	Yes			
523	✓		in A_in[2]	Location	PIN_AC27	Yes			
524	✓		in A_in[1]	Location	PIN_AC28	Yes			
525	✓		in A_in[0]	Location	PIN_AB28	Yes			
526		<<new>>	<<new>>	<<new>>					

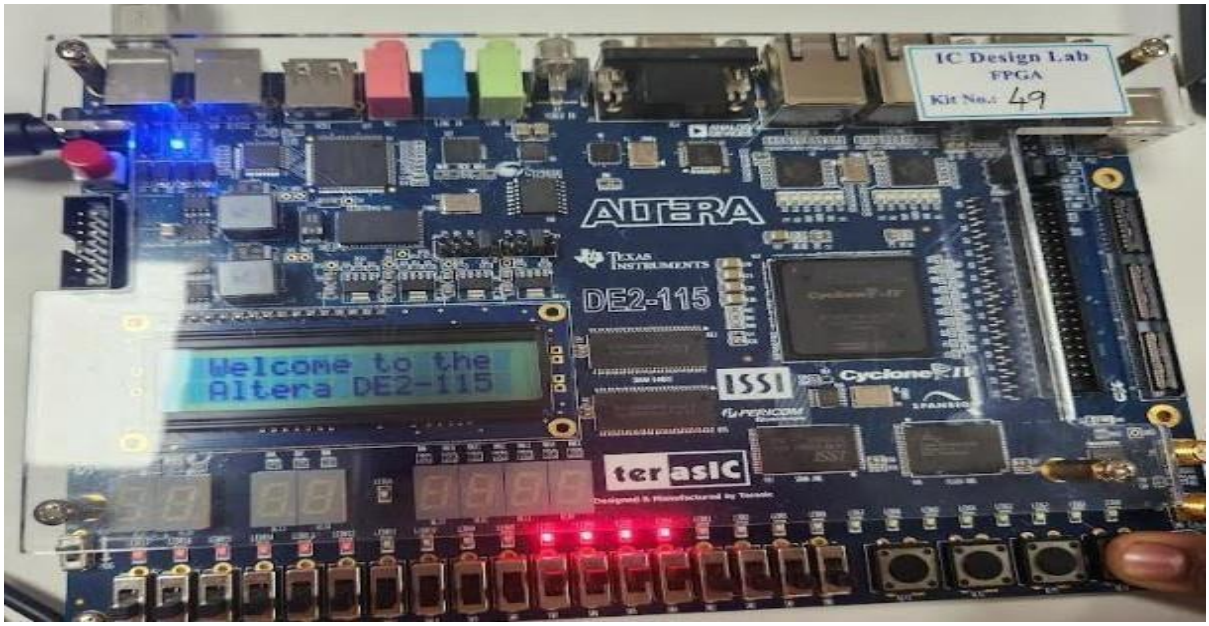
This cell shows the status of the assignment in the current row

FPGA BOARD IMPLEMENTATION

A

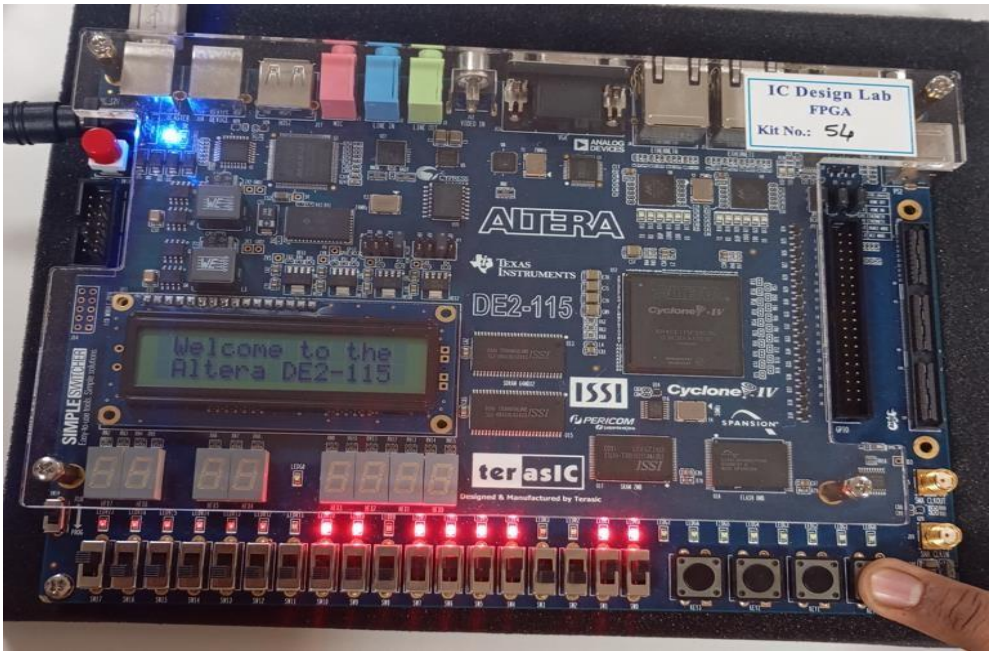


B

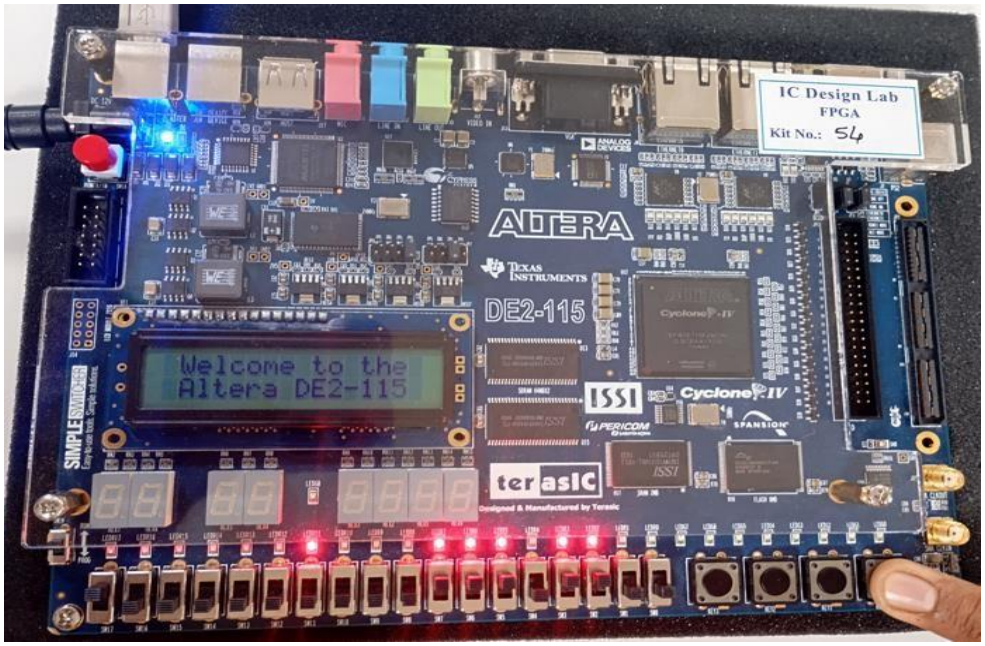


1. if (A < B) ➡ Shift A left by 1 and transfer to C

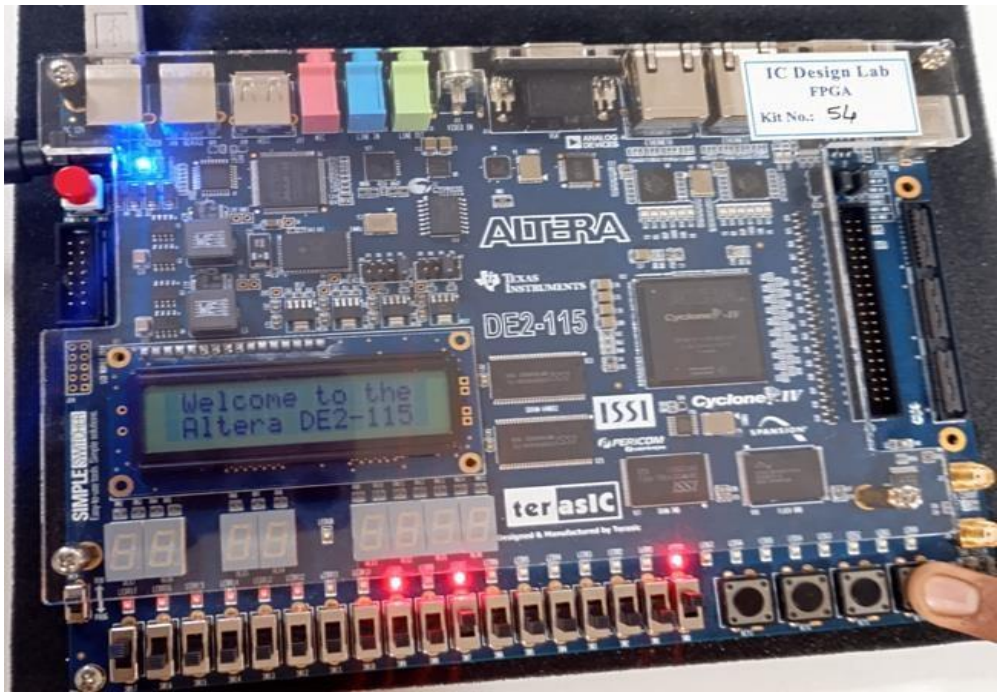
A (0011) B (1111) = C (0110)



A (1100) B (1110) = C (1000)

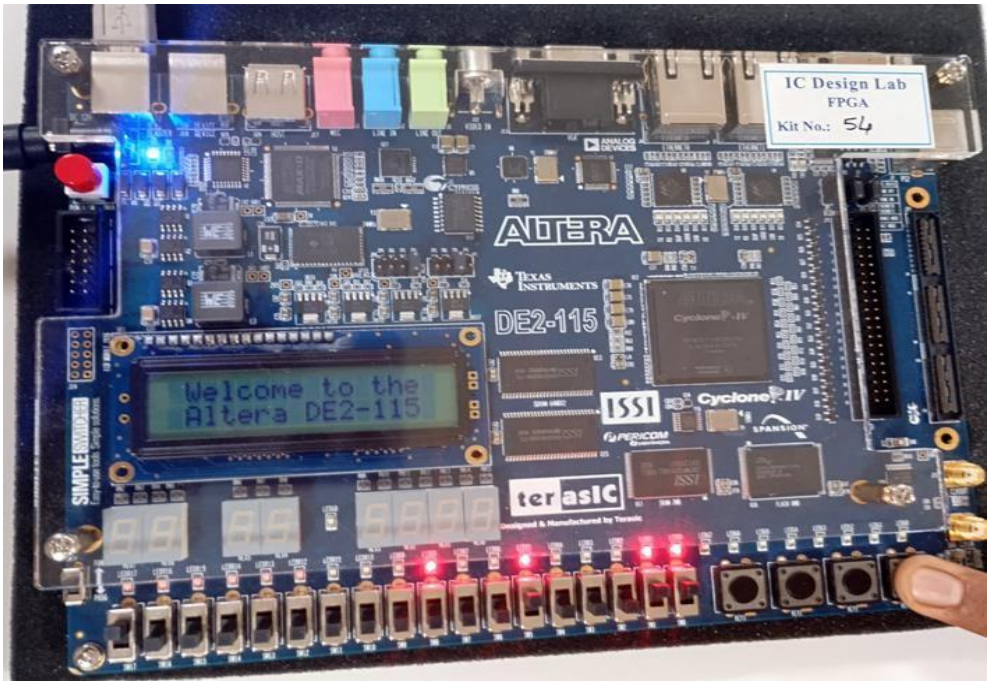


A (0001) B (1000) = C (0010)

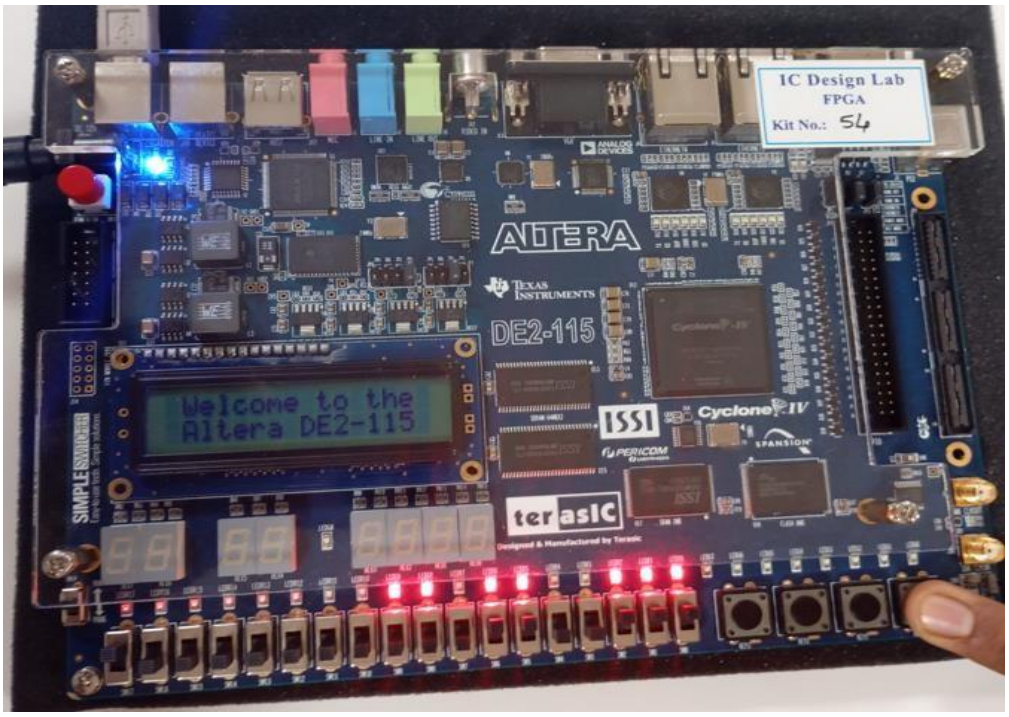


2. if (A > B) ➡ Shift B right by 1 and transfer to C

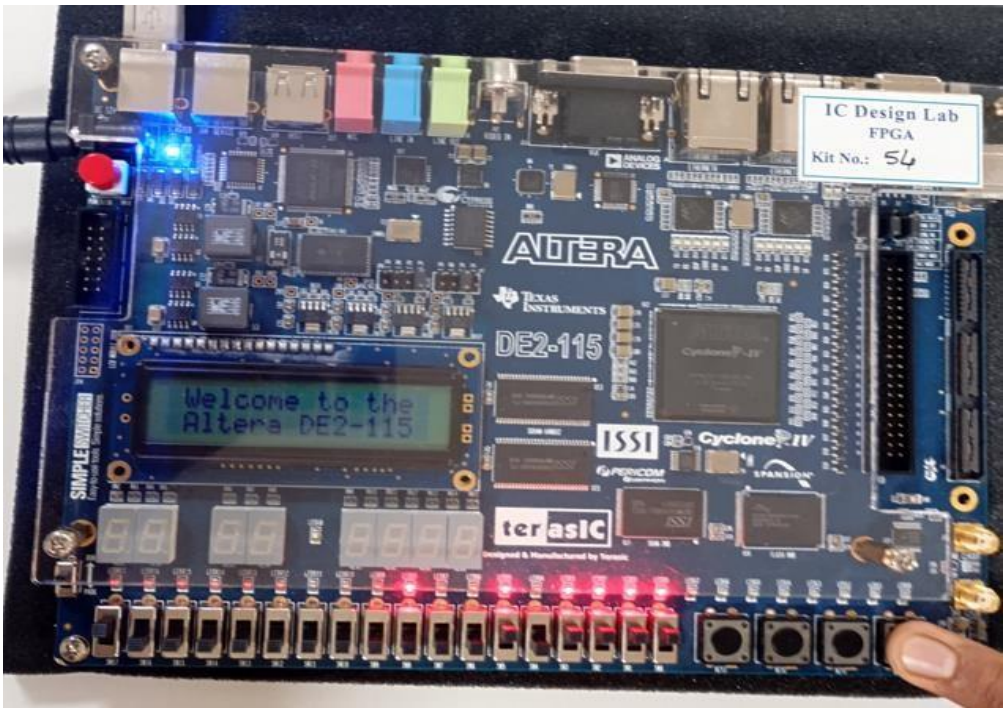
A (0011) B (0010) = C (0001)



A (0111) B (0110) = C (0011)

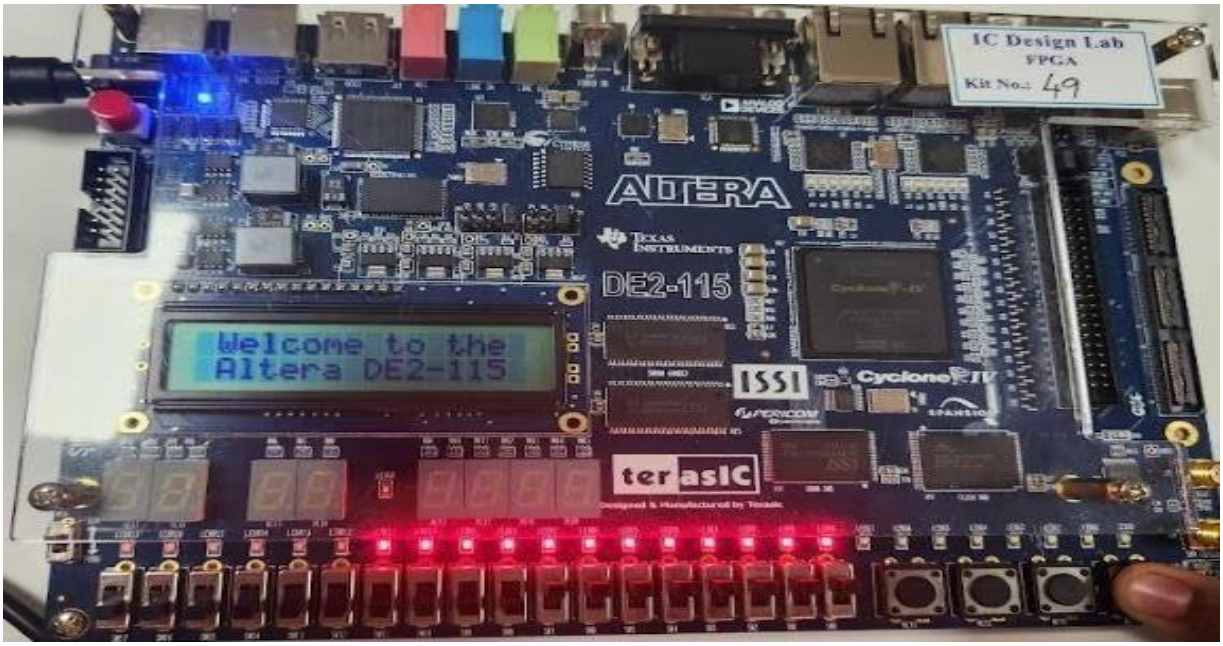


A (1111) B (0010) = C (0001)



3. if (A=B) ⟹ Transfer A to C

A (1111) B (1111) = C (1111)



A (1000) B (1000) = C (1000)

